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Genome-wide identification of *GhEDS1* gene family members in cotton and expression analysis in response to biotic and abiotic stresses

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Abstract

Background Enhanced Disease Susceptibility 1 (EDS1) genes are central regulators of plant immunity and abiotic stress responses. Although well studied in model species, their genome-wide characterisation in cotton (*Gossypium* spp.) remains lacking.

Results We identified 268 putative *EDS1* genes across four Gossypium species (*G. hirsutum*, *G. barbadense*, *G. arboreum*, and *G. raimondii*) using HMMER-based domain searches. Phylogenetic analysis grouped the genes into five subfamilies, reflecting both conserved ancestry and subgenome-specific diversification. Chromosomal mapping, collinearity, and Ka/Ks analyses revealed that segmental and whole-genome duplications were the primary drivers of expansion, with most duplicates under purifying selection. Promoter analysis using PlantCARE uncovered cis-regulatory elements responsive to abscisic acid, jasmonic acid, drought (MBS), and light signals (G-box). miRNA target prediction via psRNATarget revealed *ghr-miR414* as a key regulator targeting multiple GhEDS1 transcripts. Functional enrichment indicated roles in lipid metabolism and ubiquitin-mediated proteolysis. Finally, RNA-seq data and qRT-PCR confirmed that *GhEDS1A-13*, *GhEDS1D-57*, and *GhEDS1D-48* were significantly upregulated under PEG-induced drought stress, implicating them in ABA-linked stress adaptation.

Conclusions This study provides the first comprehensive characterisation of the EDS1 gene family in cotton, highlighting its evolutionary dynamics, regulatory complexity, and potential in improving drought tolerance through molecular breeding.

Keywords *Gossypium hirsutum*, *EDS1* gene family, Drought stress, Gene duplication, Promoter analysis, RNA-seq, PEG stress, Functional genomics

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Introduction

Cotton (*Gossypium* spp.) is a globally important industrial crop valued for its natural fibres, oilseeds, and feed potential [1]. However, its productivity is increasingly constrained by climate variability and environmental challenges, particularly drought stress [2]. Drought severely hampers cotton growth, photosynthesis, and reproductive development, ultimately reducing yield and fibre quality [3, 4]. In arid and semi-arid regions, intensifying water scarcity aggravates osmotic imbalance, oxidative stress, and metabolic dysfunction in cotton plants [5]. Given the central role of drought in limiting productivity, deciphering the genetic and molecular basis of drought tolerance is critical for sustainable improvement [6]. Targeted identification of stress-responsive genes especially those regulating key transcriptional and signalling pathways can inform the development of drought-resilient cultivars through molecular breeding and genetic engineering [7].

Among these regulatory components, the Enhanced Disease Susceptibility 1 (EDS1) gene family plays a pivotal role in plant immune and abiotic stress signalling [8]. Together with its conserved partners Phytoalexin Deficient 4 (PAD4) and Senescence-Associated Gene 101 (SAG101), EDS1 proteins contain a characteristic α/β-hydrolase-like domain and a C-terminal EP domain unique to this complex [9]. These proteins serve as molecular hubs in the integration of immune and stress responses, forming heterodimeric complexes that mediate signal transduction under biotic and abiotic challenges [10]. In *Arabidopsis thaliana*, for instance, EDS1–PAD4 complexes modulate pathogen resistance via ADR1, while EDS1–SAG101–NRG1 modules mediate cell death and defense amplification [11].

Beyond its canonical role in immune responses, growing evidence suggests that EDS1 and its co-regulators PAD4 and LSD1 also contribute to abiotic stress adaptation [12]. In *Arabidopsis thaliana*, the EDS1–PAD4 module enhances drought tolerance by modulating salicylic acid (SA)-dependent signalling, regulating developmental trade-offs and nutrient allocation [13]. Under hypoxia, the EDS1–PAD4–LSD1 complex promotes lysigenous aerenchyma formation via H_2O_2 and ethylene signalling, facilitating internal gas exchange [14]. In cereal crops such as *Triticum aestivum* and *Oryza sativa*, EDS1 homologues have been shown to improve drought and heat tolerance by maintaining redox homeostasis and enhancing antioxidant enzyme activity [15, 16]. Similarly, in *Glycine max*, silencing of GmEDS1a/b and GmPAD4 leads to heightened susceptibility to *Phytophthora sojae* and soybean mosaic virus due to impaired SA accumulation [17]. Induction of EDS1 expression upon pathogen attack has also been reported in *Vitis vinifera*, reinforcing the gene's multifunctionality [17]. These studies collectively underscore the conserved and dual functionality of EDS1 in mediating both abiotic and biotic stress responses through redox modulation, hormonal cross-talk, and signalling integration.

Despite the pivotal role of EDS1 genes in plant immunity and abiotic stress regulation, their genome-wide characterisation in cotton has not been previously reported. This gap is particularly significant given the complexity of the *Gossypium hirsutum* polyploid genome and the increasing exposure of cotton crops to drought, salinity, and viral threats under climate variability [18–21]. Moreover, recent field-based studies highlight the pressing need to enhance stress resilience in cotton breeding programmes [22]. To address this, we conducted the first genome-wide identification and comparative characterisation of the EDS1 gene family across four *Gossypium* species: the cultivated allotetraploids *G. hirsutum* and *G. barbadense*, and their diploid progenitors *G. arboreum* (A genome) and *G. raimondii* (D genome). Including the diploid lineages enabled us to trace the evolutionary origin, expansion, and divergence of EDS1 genes prior to polyploidisation, while analyses in the tetraploids revealed patterns of gene retention, duplication, and potential neofunctionalisation. We systematically identified EDS1 family members and analysed their phylogeny, gene structures, conserved domains, promoter elements, chromosomal localisation, and expression profiles under abiotic stress conditions.

Materials and methods

EDS1 gene family identification

The annotation and reference genome files for *Gossypium barbadense* (ZJU-V1.1) and *G. hirsutum* (TM1-ZJU-V2.1) (tetraploid cotton species), and their diploid progenitors *G. raimondii* (BGI) and *G. arboreum* (JGI), were retrieved from the CottonMD database (https://yanglab.hzau.edu.cn/CottonMD/download.1, accessed on 10 March 2025) [23]. The *Arabidopsis thaliana* EDS1 (AtEDS1) protein sequence was obtained from TAIR (https://www.arabidopsis.org/, accessed on 12 March 2025) and used as a reference. Characteristic EDS1 domains, namely the lipase-3 domain (PF01764) and the EP domain (PF18117), were confirmed using the Pfam database (https://pfam.xfam.org/, Pfam release 35.0, accessed on 13 March 2025) and InterPro (https://www.ebi.ac.uk/interpro/, InterPro version 96.0, accessed on 13 March 2025) [24, 25]. The GEDS1 protein domain sequences were extracted from the protein data of four cotton species using BLASTP alignment and HMMER (v3.3.2) software. To verify the potential protein sequences, the E-value threshold was set at 1e-20. To ensure higher confidence, the identified candidate proteins were further confirmed using Pfam (release 35.0), the SMART database (http://smart.embl-heidelberg.de/, accessed on

14 March 2025) [26], and the NCBI Conserved Domain Database (CDD) (https://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi, accessed on 14 March 2025) [27].

Analysis of physicochemical properties and prediction of subcellular location

To determine the isoelectric point (pI), protein size, and molecular weight of the EDS1 proteins, physicochemical analyses were performed using the ExPASy ProtParam program (https://web.expasy.org/protparam/, accessed on 15 March 2025) [28]. In addition, the prediction of subcellular localisation of EDS1 family members was performed using WoLF PSORT (https://wolfpsort.hgc.jp/, WoLF PSORT version 0.2, accessed on 15 March 2025) [29].

Evolutionary analysis of EDS1 in cotton

The amino acid sequences of the cotton EDS1 proteins and their orthologues in *Arabidopsis thaliana*, *G. barbadense*, *G. raimondii*, rice, and *G. arboreum*, obtained from the TAIR (https://www.arabidopsis.org/, accessed on 12 March 2025) and CottonMD (https://yanglab.hzau.edu.cn/CottonMD/download.1, accessed on 10 March 2025) databases, were subjected to multiple alignment using ClustalW implemented in MEGA 11. A phylogenetic tree was then constructed using MEGA 11 (version 11.0.13, https://www.megasoftware.net/, accessed on 16 March 2025) employing the maximum likelihood method with 1000 bootstrap replications [30]. The resulting tree was visualised using Interactive Tree of Life (iTOL) (version 6.8, https://itol.embl.de/, accessed on 16 March 2025).

Chromosomal location, collinearity, duplication, and Ka/Ks value analysis

The precise chromosomal locations of each EDS1 gene in the cotton species genomes were identified using annotation files obtained from the Cotton Database Resources (CottonMD; https://yanglab.hzau.edu.cn/CottonMD/download.1, accessed on 10 March 2025). Chromosomal mapping was subsequently performed using the MG2C (MapGene2Chromosome) visualisation tool, version 2.1 (http://mg2c.iask.in/mg2c_v2.1/, accessed on 17 March 2025) [31]. The syntenic relationships among EDS1 genes were analysed using the MCScan**X** (Multiple Collinearity Scan) module integrated within TBtools software, version 1.108 (https://github.com/CJ-Chen/TBtools, accessed on 17 March 2025) [32]. Collinearity maps were generated using the Basic CIRCOS function in TBtools (v1.108). Synonymous (Ks) and non-synonymous (Ka) substitution rates were also calculated using the Ka/Ks calculator embedded in TBtools v1.108, with Ka/Ks values interpreted as follows: <1 indicates purifying selection, >1 indicates positive selection, and =1 indicates neutral evolution.

Analysis of conserved motifs, domains, and exon–intron arrangement

To analyse the conserved motifs and domains of EDS1 proteins, we employed the Multiple Expectation Maximisation for Motif Elicitation (MEME) online tool (https://meme-suite.org, version 5.5.1, accessed on 18 March 2025), using default parameters, with the maximum number of motifs set to 10 and the optimal motif width defined between 6 and 50 amino acids. Identified motifs were manually inspected to assess their conservation across family members and their potential functional relevance. The amino acid sequences of EDS1 proteins were further examined using the Conserved Domain Database (CDD) (https://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi, accessed on 20 March 2025) with an E-value threshold of 1.0e − 5 to ensure statistically robust domain predictions.

Moreover, TBtools software (version 1.108) [32] (https://github.com/CJ-Chen/TBtools, accessed on 17 March 2025) was used to generate gene structure diagrams illustrating the exon–intron organisation of EDS1 genes. Gene annotation files in GFF3 format were retrieved from the CottonMD database (accessed on 22 March 2025), and exon–intron structures were visualised using the "Gene Structure View" function in TBtools, with exon lengths, intron sizes, and intron phases retained to accurately represent genomic architecture.

Cis-Acting element analysis in *GhEDS1* promoters

To investigate potential regulatory elements involved in the transcriptional regulation of *EDS1* genes, promoter sequences spanning 2,000 base pairs upstream of the translational start site were retrieved from the *Gossypium hirsutum* genome using gene annotation data from the CottonMD database (https://yanglab.hzau.edu.cn/CottonMD/, accessed on 25 March 2025). The prediction of cis-acting regulatory elements within these promoter regions was conducted using the PlantCARE database (https://bioinformatics.psb.ugent.be/webtools/plantcare/html/, accessed on 28 March 2025). Core promoter elements (TATA-box and CAAT-box) were excluded from downstream analysis. The presence of hormone-, light-, stress-, and development-related elements was summarised and visualised using TBtools (version 1.108) (https://github.com/CJ-Chen/TBtools, accessed on 27 March 2025) [33].

Prediction of putative MiRNAs targeting *EDS1* genes and enrichment analyses

To identify potential microRNAs (miRNAs) regulating the *EDS1* gene family, mature cotton miRNA sequences reported in previous studies were obtained [34]. The coding sequences (CDS) of all identified *EDS1* genes were submitted to the psRNATarget server (https://www.zhaolab.org/psRNATarget/, release 2023, accessed on 19 March 2025) using default parameters.

The predicted miRNA–target interaction networks were visualised using Cytoscape software (version 3.10.1) (https://cytoscape.org/download.html, accessed on 19 March 2025). Functional enrichment analysis of *EDS1* genes was conducted via Gene Ontology (GO) annotation using AgriGO v2.0 (http://systemsbiology.cau.edu.cn/agriGOv2/, accessed on 19 March 2025), and Kyoto Encyclopaedia of Genes and Genomes (KEGG) pathway analysis via the KEGG database (https://www.genome.jp/kegg/, accessed on 19 March 2025). These analyses provided insight into the biological processes and pathways potentially influenced by miRNA-mediated regulation of *EDS1* genes.

Plant materials and treatments

'Shayan', a drought-tolerant upland cotton variety (G. *hirsutum* L.), was used in this study. The seeds were kindly provided by the Cotton Research Institute of Iran (CRII), Gorgan, Iran. Plump and uniformly sized seeds were selected and sown in heat-sterilised sand pots. The plants were grown under controlled conditions at 28 °C during the day and 25 °C at night, a 16-hour lightand 8-hour dark period and a relative humidity of 75. After full development of the third true leaf, the plants were divided into two groups: The experimental group was treated with 20% polyethylene glycol (PEG-6000) to simulate drought stress, while the control group was irrigated regularly with distilled water. Leaf tissue was collected from both groups 24 h after treatment, immediately frozen in liquid nitrogen and stored at − 80 °C for subsequent RNA extraction and qRT-PCR analysis.

RNA isolation and quantitative Real-Time PCR (RT-qPCR)

Total RNA was extracted from frozen leaf tissues using the Hotborate method [35]. First-strand cDNA synthesis was performed using the HiScript III Reverse Transcription Kit (Qiagen, Germany) following the manufacturer's protocol. Quantitative real-time PCR was conducted on an ABI 7500 Fast Real-Time PCR System (Thermo Fisher Scientific, MA, USA) using Taq Pro Universal SYBR qPCR Master Mix (Qiagen, Hilden, Germany). Each reaction included three biological replicates. Relative expression levels were calculated using the $2^{-\Delta\Delta Ct}$ method, with *GhUBQ7* employed as the internal reference gene. Primer sequences used in this study are listed in Table S1.

Results

Characterization and molecular analysis of cotton EDS1 proteins

In this study, we identified a total of 268 EDS1 genes across four *Gossypium* species: *G. hirsutum* (89 genes), *G. barbadense* (86), *G. arboreum* (48), and *G. raimondii* (45). Homologues were identified based on the presence of conserved lipase-3 (PF01764) and EP (PF18117) domains using *Arabidopsis thaliana* EDS1 proteins as queries. An overview of gene counts and predicted physicochemical properties is provided in Table 1.

The cumulative number of EDS1 genes in the diploid progenitors (93) slightly exceeds that in the tetraploid species, possibly reflecting gene loss, subfunctionalisation, or differential retention following allopolyploidisation. The marginally higher gene count in *G. hirsutum* compared to *G. barbadense* may be due to lineage-specific expansion or annotation artefacts. Physicochemical analysis revealed that tetraploid species had comparable protein length, isoelectric point, and molecular weight distributions, while the diploid genomes exhibited greater variability, particularly in protein size and hydrophilicity. Despite this, GRAVY scores across all species were negative, indicating a consistent hydrophilic nature of EDS1 proteins. Similarly, aliphatic and instability indices suggested a generally conserved thermostability and stability profile across the family.

Subcellular localisation predictions using CELLO indicated that most EDS1 proteins are likely localised in the cytoplasm and chloroplasts, with fewer predicted in the nucleus or mitochondria (Figure S1, Table S2). This pattern is consistent with their roles in intracellular signalling and immune regulation.

Evolutionary relationships and phylogenetic classification of *EDS1* genes in *Gossypium*

The phylogenetic tree constructed from the EDS1 protein sequences of four cotton species (G. *hirsutum*, G. *barbadense*, G. *arboreum* and G. *raimondii*) and *Arabidopsis thaliana* groups all 272 sequences into six distinct lineages, labelled GEDS1-I to GEDS1-VI (Fig. 1).

Table 1 Summary of EDS1 gene characteristics in four *Gossypium* species**.** The table shows gene counts and ranges of protein length, molecular weight (MW), isoelectric point (pI), and GRAVY values for EDS1 proteins identified in each species

Species	Genome Type	No. of EDS1 Genes	Protein Length (aa)	Molecular Weight (kDa)	pI Range	GRAVY Range
G. hirsutum	Allotetraploid	89	172–830	19.95–92.13	5.01–9.88	−0.544 to − 0.030
G. barbadense	Allotetraploid	86	239–830	26.36–92.13	5.02–9.88	−0.535 to − 0.028
G. arboreum	Diploid (A)	48	356–1442	38.36–164.99	5.08–9.71	−0.676 to − 0.030
G. raimondii	Diploid (D)	45	350–1442	36.62–164.99	5.08–9.68	−0.657 to − 0.027

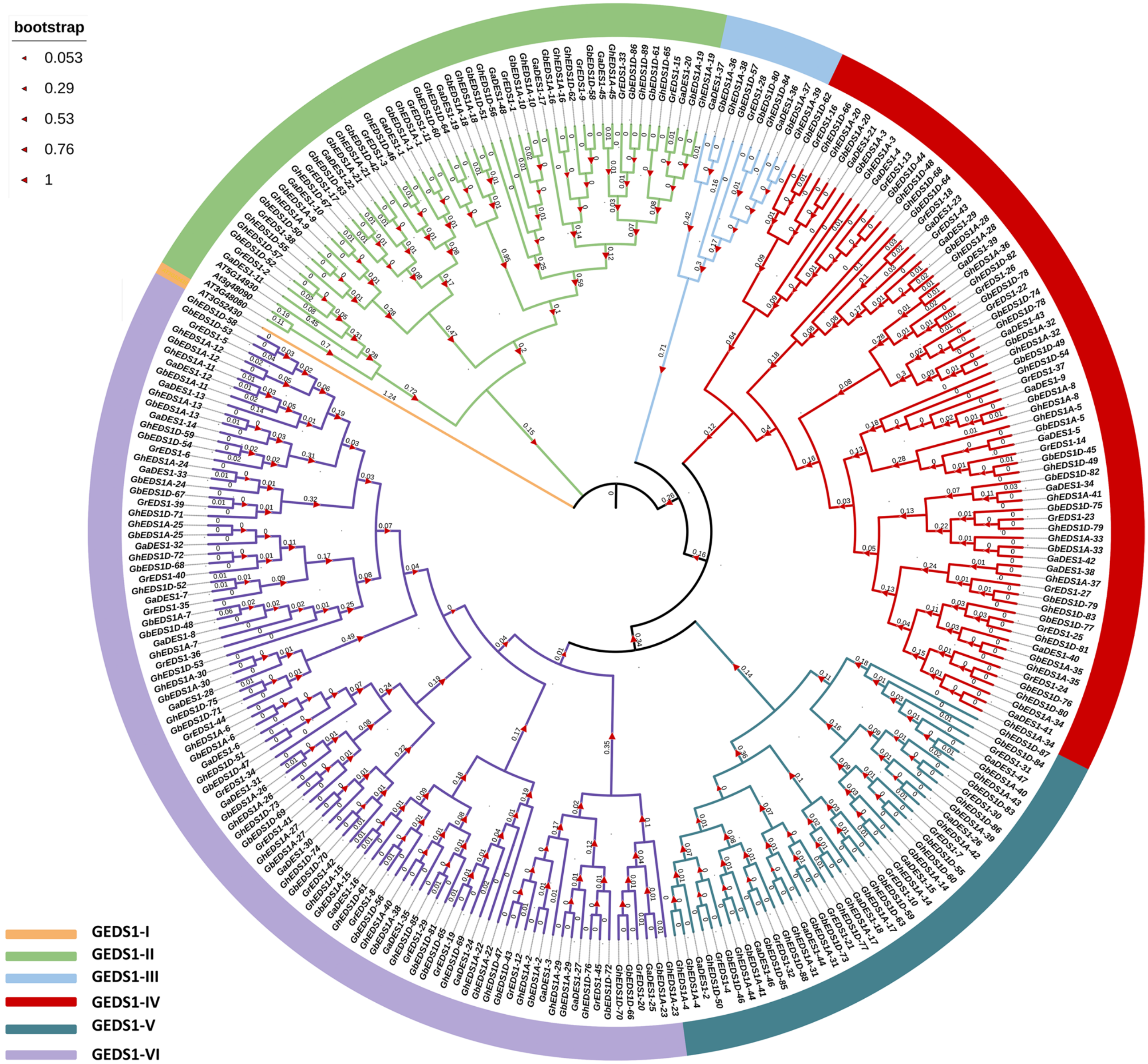

Fig. 1 Phylogenetic analysis of EDS1 genes from *Arabidopsis thaliana* and four cotton species. The tree was constructed using MEGA 11 with the maximum likelihood algorithm and 1,000 bootstrap replicates. Prefixes Ga, Gr, Gh, and Gb represent *G. arboreum*, *G. raimondii*, *G. hirsutum*, and *G. barbadense*, respectively

The distribution of EDS1 genes among these lineages was uneven, with GEDS1-IV containing the most genes (78), followed by GEDS1-I (62) and GEDS1-VI (54). The GEDS1-II and GEDS1-III lines contained the fewest members with 21 and 17 genes, respectively.

Most lines contained genes from both diploid (A and D genomes) and tetraploid cotton species, supporting the hypothesis that the tetraploid Gossypium species have retained a considerable proportion of EDS1 genes from their diploid ancestors. In many clades, genes from the A-subgenome and the D-subgenome of tetraploid cotton were close to each other, suggesting conserved homologous gene relationships. For example, within GEDS1-I and GEDS1-IV, numerous gene pairs from *G. hirsutum* and *G. barbadense* were found alongside their orthologues in *G. arboreum* and *G. raimondii*, suggesting common ancestry and the conservation of syntenic gene blocks after polyploidisation. Remarkably, three of the four EDS1 proteins from *Arabidopsis thaliana* belong to the large cotton clades GEDS1-I, GEDS1-IV and GEDS1-V, indicating conserved evolutionary relationships between cotton and other dicotyledons. However, an EDS1 gene from *A. thaliana* formed a specific clade without any cotton EDS1. The presence of both diploid and tetraploid cotton genes in almost all lines is also consistent with previous studies showing that the A and D genomes of tetraploid cotton arose from ancient hybridisation and genome duplication events. An exception was observed in GEDS1-III, where a subgroup consisted exclusively of A-subgenome-derived genes, indicating a possible subgenome-specific expansion or differential gene loss in the D-subgenome. These phylogenetic results demonstrate that the *EDS1* gene family in *Gossypium* is characterized by both conserved ancestry and genome-specific diversification, providing a valuable framework for understanding gene function and evolution in the context of the complex polyploid history of cotton.

Chromosomal distribution, multiple synteny, gene duplication, and collinearity analysis of *GhEDS1* genes

Chromosomal mapping revealed an uneven distribution across subgenomes and chromosomes (Fig. 2). In *G. hirsutum*, 45 EDS1 genes were located on subgenome A and 44 on subgenome D. Chromosomes A11, A08 and A10 harboured the highest number of genes (11, 8 and 7, respectively), which is reflected in D11, D08 and D10. *G. barbadense* showed a similar enrichment on A08, A10 and D11, while A01, A06, D01 and D06 were sparsely represented (with only one gene each).

In *G. arboreum*, the EDS1 genes were distributed on 10 chromosomes and one unanchored scaffold, with the highest densities observed on chromosomes A11 (10 genes), A10 (8), and A08 (7). No genes were localised on chromosomes A04, A05, or A07, explaining the absence of EDS1 representation on three chromosomes in Fig. 2C. In *G. raimondii*, EDS1 genes were mapped onto 11 chromosomes, with D04, D06, and D11 harbouring the most members, while D13 and D05 lacked any EDS1 loci. The recurrent accumulation of *EDS1* genes on chromosomes 08, 10, and 11 across both diploid and tetraploid genomes suggests ancestral hotspots of expansion. In contrast, sparsely populated chromosomes may reflect local gene loss or suppressed duplication events. These findings offer a genome-wide perspective on the chromosomal organisation and evolutionary dynamics of the EDS1 gene family in *Gossypium* species (Fig. 2).

To investigate the evolutionary conservation and chromosomal localization of GhEDS1 genes in related species, an interspecific synteny analysis was performed using the MCScan algorithm of the JCVI toolkit. The resulting collinearity relationships were visualized between *G. hirsutum* and four reference genomes: *G. arboreum*, *G. raimondii*, *G. barbadense* and *Arabidopsis thaliana* (Fig. 3A). A total of 35, 35, 82 and 27 syntenic gene pairs were identified between *G. hirsutum* and *G. arboreum*, *G. raimondii*, *G. barbadense* and *A. thaliana*, respectively (Table S3a). The largest number of collinear EDS1 orthologues was observed between *G. hirsutum* and *G. barbadense*, distributed across both the A and D subgenomes, reflecting the close evolutionary relationship between these tetraploid species. In contrast, the syntenic gene pairs between *G. hirsutum* and its diploid ancestors *G. arboreum* (A genome) and *G. raimondii* (D genome) were more clearly separated. The orthologues of *G. arboreum* were predominantly aligned with the A subgenome of *G. hirsutum*, whereas those of *G. raimondii* were mainly associated with the D subgenome. This subgenomic correspondence allowed the preliminary assignment of 11 GhEDS1 genes to each subgenome based on exclusive, non-overlapping syntenic matches. Comparative analysis with *A. thaliana* revealed fewer conserved syntenic blocks spanning across multiple chromosomes with low linkage density. This observation is consistent with the greater phylogenetic distance and genomic divergence between *A. thaliana* and Gossypium species. Taken together, these results delineate the orthologous relationships and subgenomic lineage of the GhEDS1 loci in the context of the diploid progenitors and closely related tetraploid genomes, as shown in Fig. 3A.

To investigate the duplication mechanisms contributing to the expansion of the GhEDS1 gene family in G. *hirsutum*, the duplication types of 89 identified genes were classified using a synteny-based approach (Table S4). Of these, 70 genes (78.7%) were categorised as whole-genome or segmental duplication (WGD/segmental), while 19 genes (21.3%) were classified as scattered duplicates. The gene set was further subdivided into the A and D subgenome. The A subgenome (GhEDS1A) contained 46 genes, of which 34 (73.9%) were WGD/segmental and 12 (26.1%) were dispersed. The D subgenome (GhEDS1D) contained 43 genes, of which 36 (83.7%) were WGD/segmental and 7 (16.3%) were dispersed duplicates. The proportion of WGD/segmental duplicates was higher in the D subgenome than in the A subgenome. In contrast, the A subgenome had a slightly higher proportion of dispersed duplicates. These proportions reflect the differences in the distribution of duplication types between the subgenomes.

Analysis of intragenomic collinearity using a Circos plot (Fig. 3B) revealed 13 paralogous gene pairs in *G. hirsutum*, with the connections distributed between A and D subgenomes. In particular, the segmentally duplicated gene pairs were predominantly located on homologous chromosomes such as A08/D08, A13/D13 and A07/D07. Red colored connections represent gene pairs under positive selection ($Ka/Ks > 1$), while blue lines indicate paralogues under purifying selection. To assess the selection pressure acting on duplicated GhEDS1 genes, the non-synonymous (Ka) and synonymous (Ks) substitution rates were calculated for 49 syntenic gene pairs (Table S3). All pairs had Ka/Ks ratios well below 1.0, indicating that the gene family evolved predominantly under purifying selection, with no evidence of recent positive selection among the paralogues analyzed. Of the 49

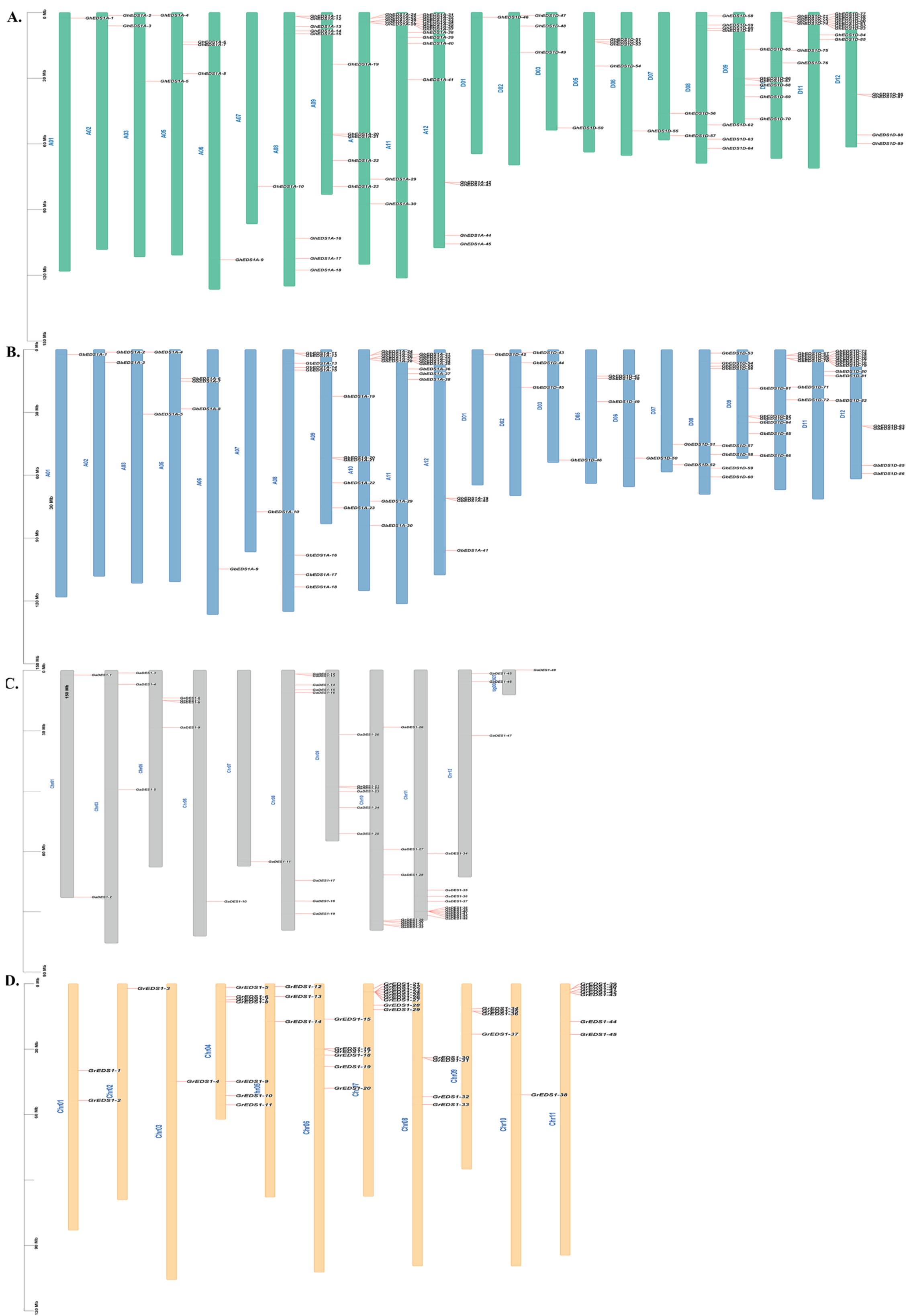

Fig. 2 Chromosomal distribution of EDS1 genes in four *Gossypium* species. Segmentally and tandemly duplicated genes are connected by lines. **A** *G. hirsutum*, **B** *G. barbadense*, **C** *G. arboreum*, **D** *G. raimondii*

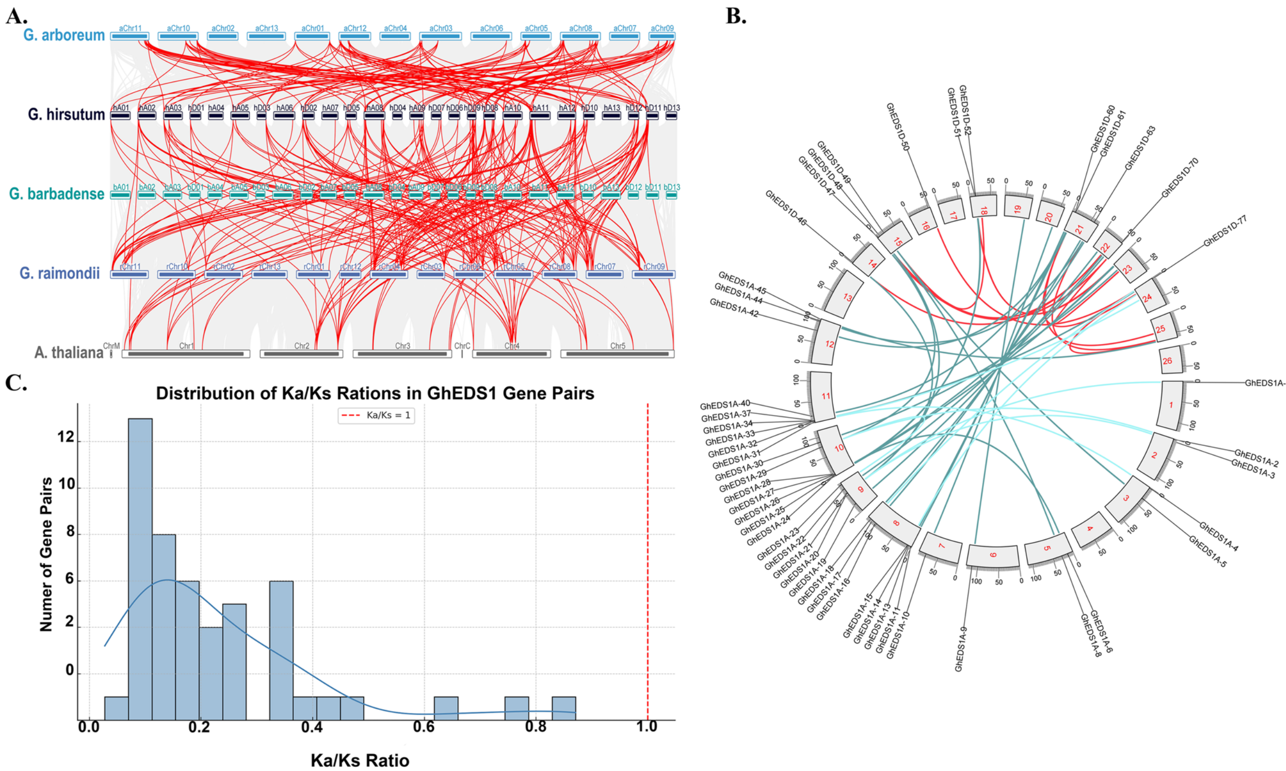

Fig. 3 Synteny and selection analysis of *G. hirsutum* EDS1 genes. **A** Interspecies synteny between *G. hirsutum* and its relatives (*G. arboreum*, *G. raimondii*, *G. barbadense*, and *A. thaliana*). **B** Intragenomic collinearity of EDS1 gene pairs in *G. hirsutum* using a Circos plot. Red and blue lines indicate positive and purifying selection, respectively. **C** Ka/Ks distribution of 89 syntenic gene pairs, showing predominance of purifying selection

gene pairs, 43 had Ka/Ks ratios below 0.5, suggesting that these duplicates are under strong functional constraint and likely retain essential functions of their ancestors. The remaining six pairs had Ka/Ks ratios between 0.5 and 0.87, with the highest observed value being 0.8707 for the pair GH_A08G0200/GH_D08G0209, indicating a moderate relaxation of selection pressure, although not yet reaching levels consistent with adaptive evolution. These results reflect a conserved evolutionary pathway for the GhEDS1 gene family in which most paralogues have fewer non-synonymous changes compared to synonymous substitutions. Such patterns are consistent with a scenario in which gene duplication was followed by sequence conservation rather than neofunctionalisation or adaptive divergence. The narrow range of Ka/Ks ratios also suggests that post-duplication diversification was limited and the functional integrity of key defense-related metabolic pathways was maintained in *G. hirsutum*.

Exploring the evolutionary relationships, conserved motifs, protein domains, and gene structures of *EDS1* Family members in G. *hirsutum*

To investigate the structural and evolutionary features of the EDS1 gene family in *G. hirsutum*, a comprehensive analysis, including phylogenetic inference, exon–intron structure, domain composition, and conserved motif patterns was performed (Fig. 4A–E). The phylogenetic tree constructed from 89 full-length GhEDS1 protein sequences (Fig. 4A) grouped the proteins into four distinct clades, each supported by robust bootstrap values. This classification provided a framework for subsequent structural and functional comparisons. The clade-specific clustering suggests potential sub-functionalization among the paralogues, possibly reflecting divergence in regulatory or biochemical roles.

Gene structure analysis (Fig. 4B) showed that exon–intron organization within clades was generally conserved. Subfamily IV members consistently possessed 11 exons, reflecting strong evolutionary conservation. The members of subfamily IV consistently possessed 11 exons, indicating strong evolutionary conservation. In contrast, subfamily I exhibited considerable variation, ranging from 3 exons (e.g., GhEDS1-09 A) to 12 exons (e.g., GhEDS1-01D), suggesting ongoing structural divergence possibly associated with functional diversification. This structural heterogeneity within subfamily I could indicate exon gain/loss events or intragenomic shuffling. Protein domain analysis using Pfam annotation (Fig. 4C) showed that most GhEDS1 proteins have the characteristic α/β-hydrolase fold (PF00561) at the N-terminus and the EP (EDS1-PAD4) domain (PF08263) at the C-terminus. These characteristic domains were conserved in

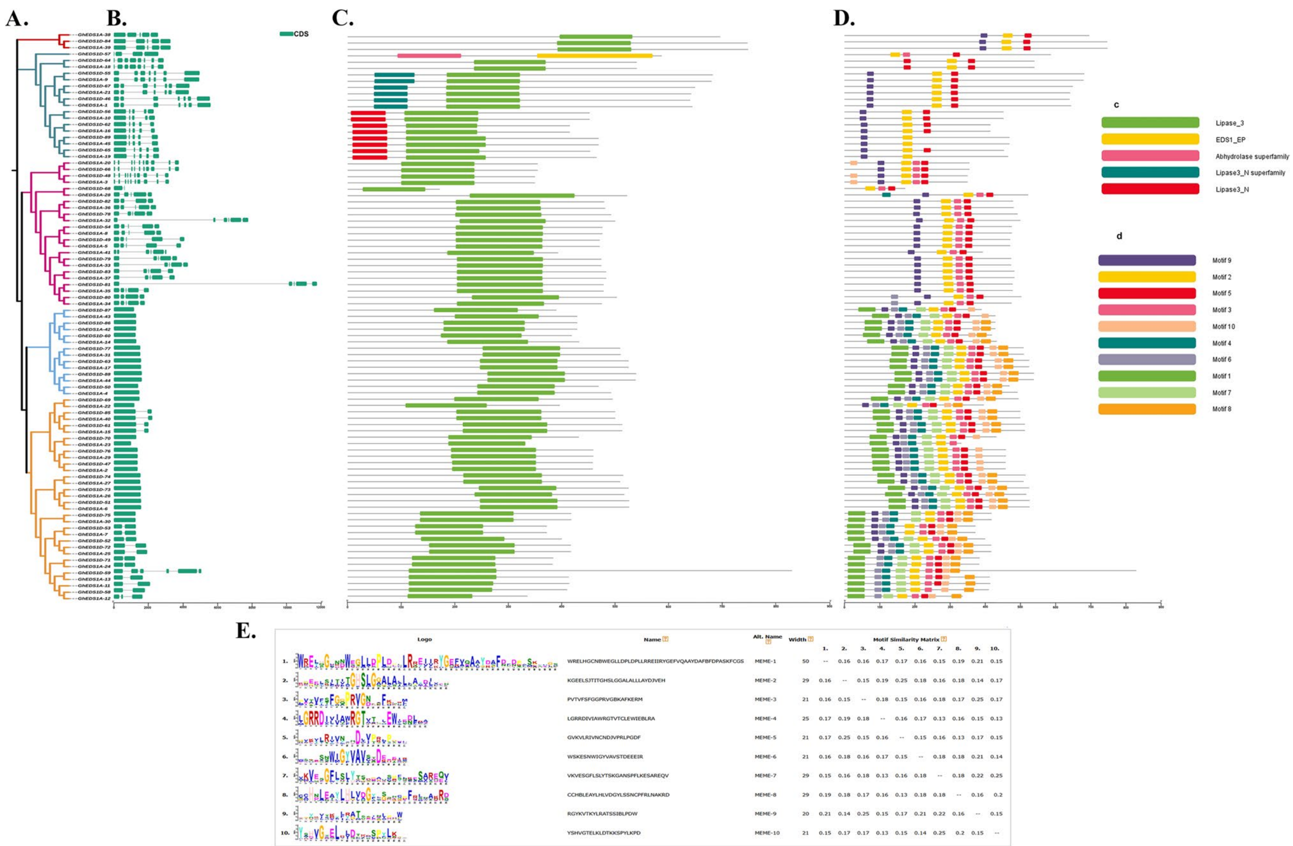

Fig. 4 Gene Structure, Conserved Motifs, Domains, and Chromosomal Localization of the GhEDS1 Gene Family in upland cotton. **A** Phylogenetic tree of *GhEDS1* genes depicting their evolutionary relationships. Different colors indicate distinct clades. **B** Domain architecture of GhEDS1 proteins, highlighting conserved functional domains with different colors. **C** Gene structure analysis displaying exon-intron organization of *GhEDS1* genes. Green boxes represent exons; black lines represent introns. **D** Conserved motif distribution among GhEDS1 proteins. Identified motifs are color-coded. **E** Sequence logo representation illustrating conserved amino acid residues in GhEDS1 proteins

most GhEDS1 members, although a few truncated proteins lacked one or both domains, indicating possible pseudogenisation or loss of function. The motif analysis performed with MEME (Fig. 4D) identified ten conserved motifs that differ in their distribution and frequency among the GhEDS1 members. Motifs 1, 2 and 3 were present in almost all proteins, indicating their essential role in maintaining the core functions of EDS1. In contrast, motif 6 and motif 9 showed clade-specific patterns. Motif 6 was absent in several members of subfamily III, suggesting a possible subfamily-specific functional divergence. These differences in the presence and arrangement of motifs support the idea of clade-specific specialisation.

Sequence logo analysis (Fig. 4E) revealed high conservation within several motifs, particularly motifs 1, 2 and 4, which have conserved amino acid residues likely involved in catalytic activity or structural stability. The similarity indices of the motifs confirmed this, with motif 1 having a high conservation value (0.83), while more variable motifs such as motif 9 had lower values (0.37). These metrics suggest that while the core structural elements of EDS1 proteins are well conserved, peripheral regions evolve under facilitating conditions, enabling functional plasticity. Taken together, these multilevel analyses show that GhEDS1 genes exhibit conserved domain and motif architecture while showing subfamily-specific structural variation, highlighting their evolutionary flexibility. These results form the basis for future studies on the functional role of individual GhEDS1 genes in stress signaling and developmental processes.

Cis-acting regulatory element analysis of GhEDS1 genes

To investigate the transcriptional regulation of *GhEDS1* genes in *G. hirsutum*, we extracted the complete 2 kb upstream promoter sequences for all 89 *GhEDS1* genes from the reference genome and analysed them using the PlantCARE database (Fig. 5A & B). The identified cis-acting elements were quantified and functionally categorised into four main groups: cell cycle and development-related, hormone-responsive, light-responsive, and stress-responsive elements, as shown in Fig. 5A (heatmap) and Fig. 5B (bar graph). The heatmap (Fig. 5A) illustrates the presence and density of individual cis-elements across GhEDS1 promoters, enabling comparison

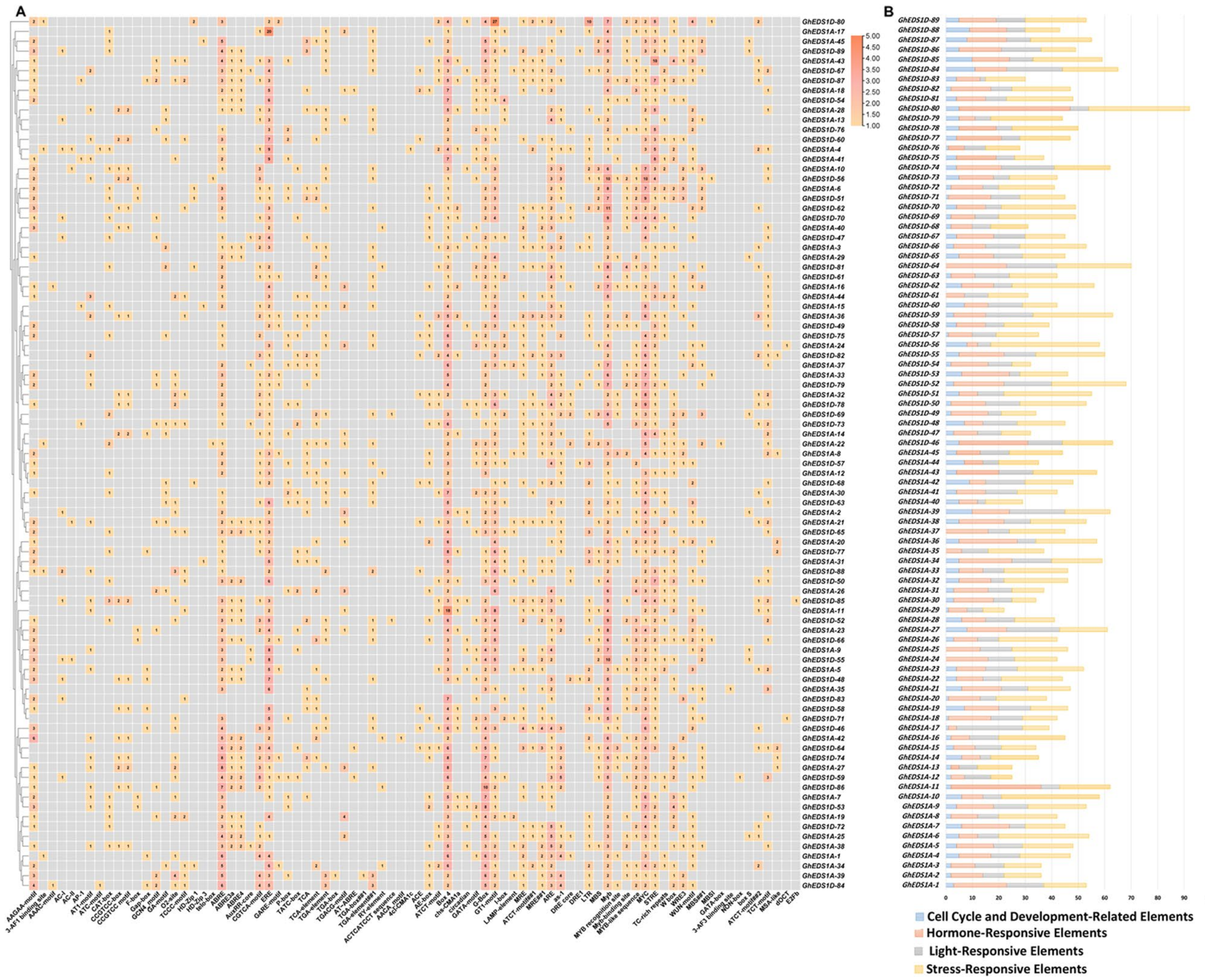

Fig. 5 **A** Heatmap illustrating the presence and distribution of cis-acting regulatory elements within the 2 kb upstream promoter regions of *GhEDS1* genes. Each cell indicates the presence of specific motif types across individual promoters, with elements functionally grouped. **B** Bar chart summarising the total number of cis-elements identified per gene, classified into four major functional categories: cell cycle and development-related, hormone-responsive, light-responsive, and stress-responsive elements

of promoter complexity and potential regulatory specialisation among gene family members.

A broad distribution of development-related motifs, such as CAT box, GCN4 motif, and O2 site, was observed across the promoters. In particular, *GhEDS1A09* and *GhEDS1D13* displayed a high abundance of developmental motifs, suggesting a possible role in tissue-specific development. Other development-associated elements, including the F-box (*GhEDS1D21*) and Gap-box (*GhEDS1A07*), were also identified, implying regulatory functions in cell cycle control and organogenesis.

Hormone-responsive elements were widespread. ABA-responsive motifs (ABRE, ABRE3a, ABRE4) were abundant in *GhEDS1D21*, *GhEDS1A09* and *GhEDS1A10*. Jasmonic acid-related elements (TGACG, CGTCA) were enriched in *GhEDS1D18* and *GhEDS1A12*, while auxin-related motifs (AuxRR-core, TGA-box) were prevalent in *GhEDS1A01* and *GhEDS1D15*. Promoters of several other genes contained motifs responsive to gibberellin (P-box, GARE motif, TATC-box), salicylic acid (TCA element), and ethylene (ERE), underscoring the complexity of hormonal regulation.

Light-responsive elements, such as Box 4, G-box and GT1 motifs, were found in almost all GhEDS1 promoters. *GhEDS1D13*, *GhEDS1A01*, *and GhEDS1A06 showed particularly high densities*, suggesting light-regulated expression patterns potentially linked to circadian and photomorphogenic responses. Stress-responsive elements were notably abundant. *GhEDS1D21*, *GhEDS1D18*, and GhEDS1A06 carried multiple drought-responsive motifs (MBS), as well as low-temperature (LTR) and dehydration-related (DRE) elements. Promoters of *GhEDS1D22*, *GhEDS1A12* and *GhEDS1D13* were enriched in biotic stress-related motifs such as WUN, W-box, and TC-rich repeats, suggesting potential involvement in pathogen defence and wounding responses. Taken together, the heat map and bar graph (Fig. 5A and B) emphasized the functional diversity of GhEDS1 regulatory landscapes, reflecting potential responsiveness to a variety of developmental, hormonal and environmental stimuli.

Protein-protein interaction and *GhEDS1* genes-transcription factor regulatory network

To investigate the potential functional associations of the *G. hirsutum* EDS1 gene family, a protein–protein interaction (PPI) network was constructed using the STRING database based on interaction knowledge deposited for *A. thaliana* (Table S5). A total of 89 GhEDS1 protein sequences were queried, yielding interactions with 23 orthologous Arabidopsis proteins, filtered using a high-confidence interaction threshold (≥0.7). The resulting network (Fig. 6A) revealed several distinct interaction clusters, indicating functional modularity between GhEDS1 proteins and their Arabidopsis-derived interactors. Notably, GhEDS1D-88 was positioned as a central node, displaying high-confidence interactions with members such as GhEDS1A-38, GhEDS1D-74, and GhEDS1D-67, as well as with lipid metabolism regulators PLA1 and PLA2-ALPHA. These associations, supported by experimental or curated evidence, suggest the involvement of this GhEDS1 subfamily in

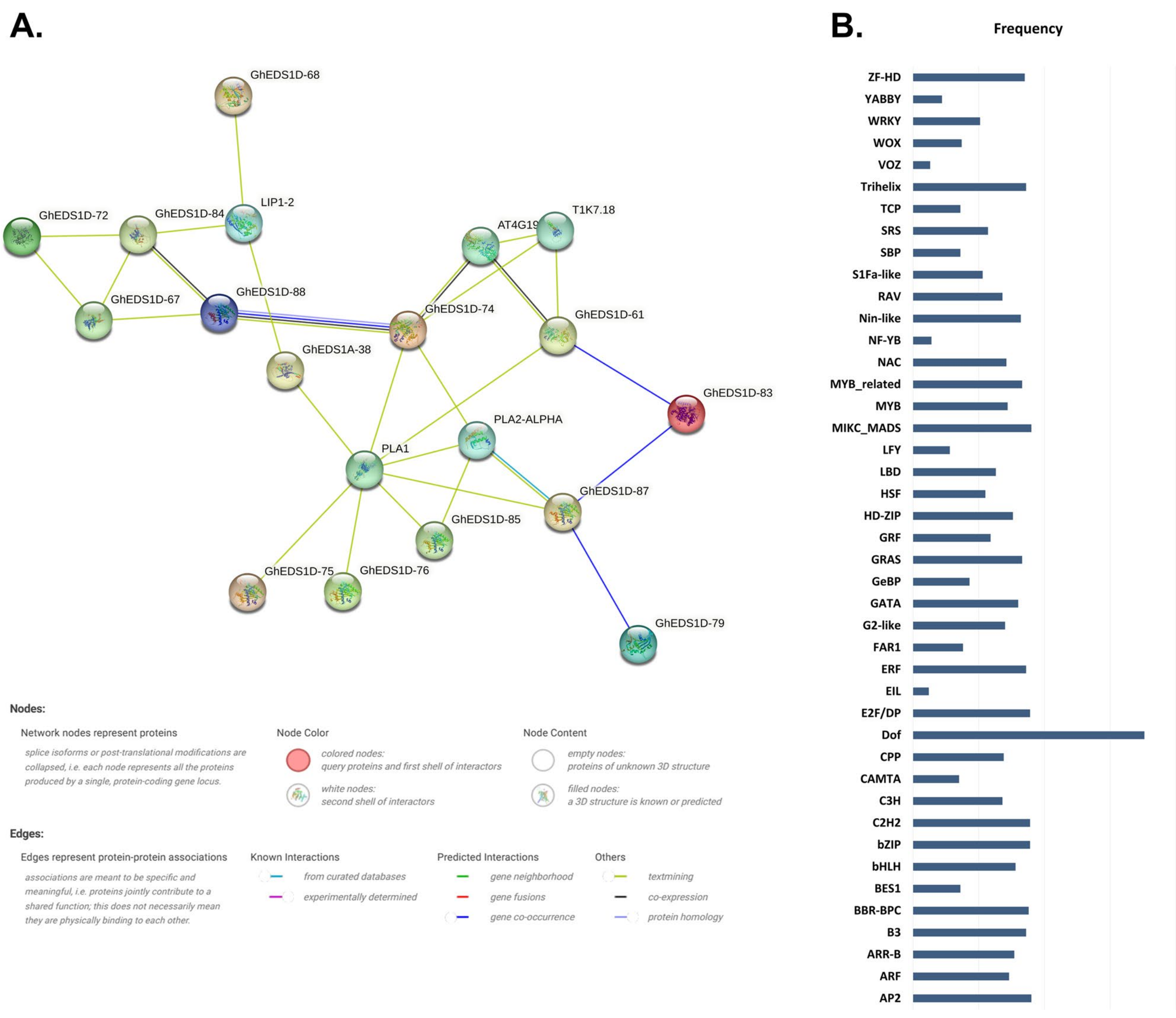

Fig. 6 Protein-protein interaction of *GhEDS1* proteins based on known Arabidopsis protein orthologous. **A** Protein–protein interaction (PPI) network of GhEDS1 proteins based on orthology with Arabidopsis thaliana, constructed using STRING (confidence ≥ 0.7). **B** Distribution of transcription factor (TF) families predicted to regulate *GhEDS1* genes based on 2,000 bp upstream sequences. Node and edge properties in (**A**) indicate interaction confidence and evidence type; top TF families are shown in (**B**)

phospholipid-related signalling pathways. Additional, GhEDS1D-72, GhEDS1D-84, and GhEDS1D-67 formed a tightly linked module, potentially reflecting conserved or redundant regulatory roles. Similarly, GhEDS1D-83, GhEDS1D-87, and GhEDS1D-79 comprised a discrete interaction cluster. Interactions with orthologues such as AT4G19520 and T1K7.18, both implicated in hormonal and immune signaling, further underscore the possible role of GhEDS1 proteins in stress-responsive regulatory networks.

Furthermore, transcription factor (TF) regulatory interactions were explored by retrieving the 2,000 bp upstream sequences of all 89 GhEDS1 genes and analyzing them via the PTRM database. This analysis identified 2,792 TFs belong to multiple families which were associated with the regulation of 87 GhEDS1 genes (Table S6). Among these, the Dof family was the most prevalent, comprising 176 TFs, followed by AP2 (90), MIKC_MADS (90), C2H2 (89), E2F/DP (88), bZIP (88), and ERF (87). By contrast, VOZ and EIL were the least represented TFs, with 13 and 1 identified TFs, respectively (Fig. 6B). All GhEDS1 genes were targeted by multiple TF families, with GhEDS1A-1 and GhEDS1D-46 each associated with 26 distinct TF families. TFs linked to abiotic stress and defense such as HD-ZIP, ARF, MYB, and NAC was prevalent. In addition, TF families involved in developmental regulation, including bHLH, TCP, WRKY, BBR-BPC, ERF, and LBD, were also identified among GhEDS1 regulatory targets.

Prediction of prospective *GhEDS1* gene-targeting MiRNAs across the genome

To investigate the post-transcriptional regulation of the GhEDS1 gene family, in silico analysis of miRNA–mRNA interactions in *G. hirsutum* was performed. This analysis identified 85 high-probability miRNA–target pairs (expected value ≤ 5.0) involving 46 different GhEDS1 genes and 32 unique miRNAs from multiple conserved families (Table S7). Among these, the ghr-miR414 family exhibited the most extensive regulatory potential. Multiple isoforms *ghr-miR414a* to *ghr-miR414f* were predicted to target *GhEDS1A-1*, *GhEDS1A-21*, *GhEDS1A-39*, *GhEDS1D-55*, *GhEDS1D-67*, and *GhEDS1D-84*, with expectation values ranging from 2.5 to 4.5. Notably, GhEDS1A-21 was co-targeted by four distinct ghr-miR414 isoforms, suggesting a potential regulatory hub within this gene family. In addition, members of the ghr-miR396 family (ghr-miR396b and ghr-miR396c) were predicted to target *GhEDS1A-36* and *GhEDS1D-82* through both mRNA cleavage and translational repression, implying a dual-layered regulatory mechanism. Other significant interactions included ghr-miR2949a–c targeting *GhEDS1A-28* and *GhEDS1D-53*, *ghr-miR2950* regulating *GhEDS1A-37* and *GhEDS1D-83*, and *ghr-miR399a–d* acting upon *GhEDS1A-30*, *GhEDS1D-75*, and *GhEDS1D-80* (Fig. 7).

Of the 85 predicted interactions, 66 (77.6%) were mediated by transcript cleavage, while 19 (22.4%) involved translational inhibition, consistent with the predominant mechanisms of miRNA action in plants. ghr-miR414d selectively targeted *GhEDS1D-19*, whereas *ghr-miR162a* modulated both *GhEDS1A-10* and *GhEDS1D-56* primarily through translational repression, highlighting the Isoform-specific and subgenome-specific regulation of the predicted regulatory pairs. These findings suggest possible subfunctionalisation among *GhEDS1* paralogues, and reflect the complexity of miRNA-mediated regulatory networks potentially modulating plant defence and stress response mechanisms.

Enrichment analyses of *GhEDS1* genes using GO and KEGG pathway analysis

To investigate the potential functional impacts of *GhEDS1* genes in *G. hirsutum*, enrichment analyses were performed using the Gene Ontology (GO) and KEGG databases. GO enrichment was analyzed in the categories of biological process (BP), molecular function (MF) and cellular component (CC) (Fig. 8A). Although none of the terms reached significance after FDR correction (q-values ≥ 0.88), several categories had low nominal p-values, providing valuable initial insights. In the BP category, the most enriched terms included 'regulation of secondary metabolic processes' (GO:0043455; $p = 4.0 \times 10\text{-}^{4}$), 'regulation of biosynthetic processes' (GO89; $p = 2.3 \times 10\text{-}^{3}$) and 'intracellular protein transmembrane import' (GO:0044743; $p = 2.4 \times 10\text{-}^{3}$), indicating a possible role in the regulation of metabolism and intracellular signaling. Enrichment for transcriptional processes such as 'regulation of gene expression' (8) and 'DNA-templated transcription' (GO:0006355) was also observed. In the MF category, GhEDS1 genes were associated with terms such as 'transcription cofactor activity' (GO:0003712) and 'transcription factor binding' (GO:0000989), while CC terms such as 'cytoplasm', 'chloroplast' and 'membrane-bound organelle' were consistent with the known subcellular localization of EDS1. Although no statistical significance was achieved, these patterns reflect known transcription- and transport-related functions of EDS1 proteins and suggest testable hypotheses for future validation.

In contrast, enrichment analysis using KEGG database revealed four significantly enriched pathways (FDR < 0.01), providing more robust evidence for functional connections of GhEDS1 (Fig. 8B). The most enriched pathway was glycerophospholipid metabolism (ghi00564), comprising 23 GhEDS1 proteins ($FDR = 1.85 \times 10^{-29}$), indicating a likely involvement in the maintenance of membrane structure and

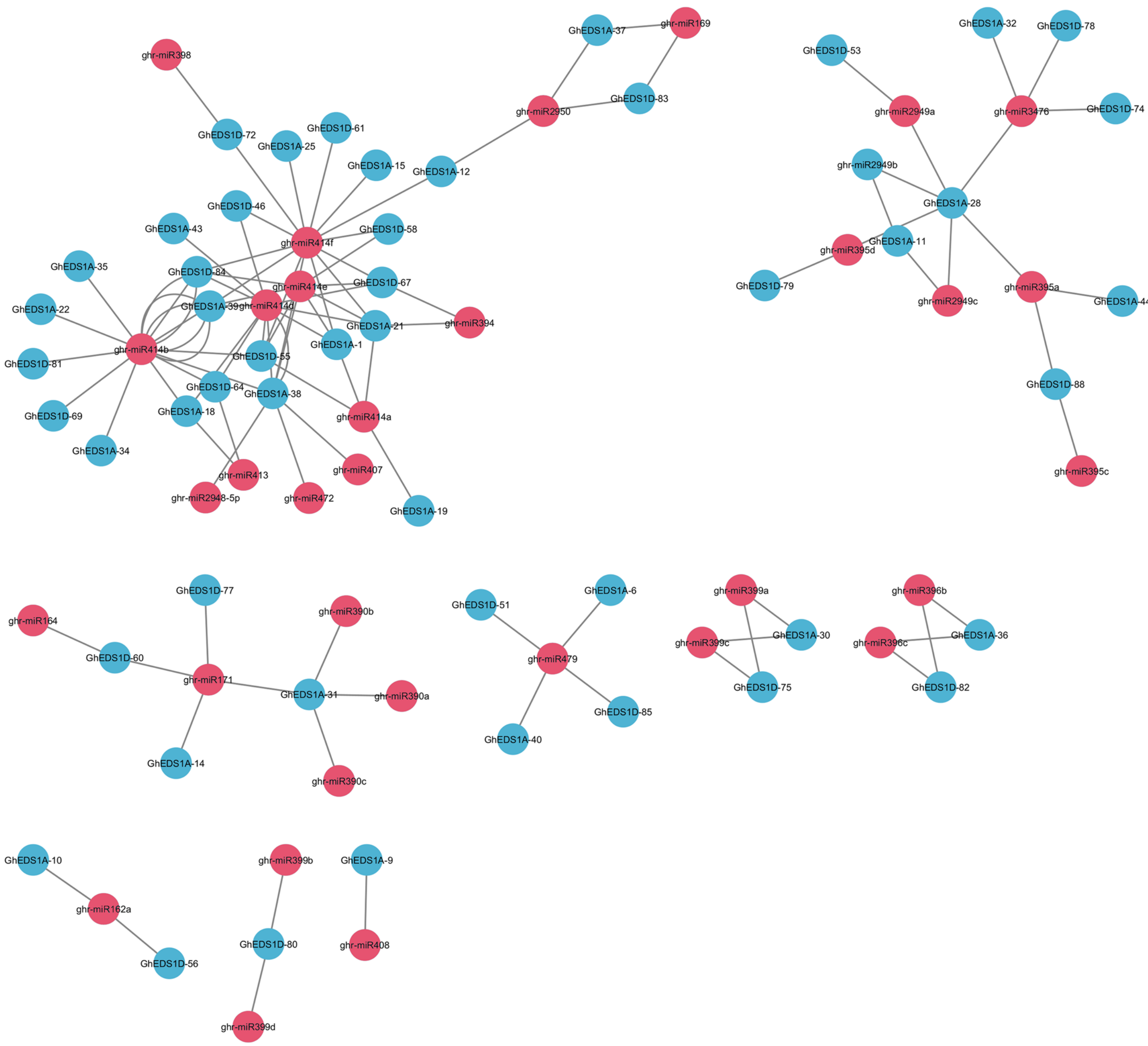

Fig. 7 Predicted miRNA–mRNA regulatory network of *GhEDS1* genes in *G. hirsutum*. Blue nodes represent *GhEDS1* genes, and circular grey nodes indicate targeting miRNAs. Arrows denote predicted regulatory interactions, including cleavage and translational repression

lipid-based signaling cascades. Nine GhEDS1 proteins were also associated with ubiquitin-mediated proteolysis (ghi04120) (FDR = 1.04×10^{-6}), indicating a possible role in the post-translational regulation of immune signaling components. Additional enrichment was noted for alpha-linolenic acid metabolism (ghi00592) (6 proteins, FDR = 6.00×10^{-6}) and ether lipid metabolism (ghi00565) (5 proteins, FDR = 1.04×10^{-5}), pathways known to influence stress resilience through jasmonic acid biosynthesis and modulation of membrane fluidity. These KEGG associations support the hypothesis that GhEDS1 proteins are involved in lipid-mediated defense mechanisms and stress-resilient regulatory networks in G. hirsutum.

Expression profiles of *GhEDS1* genes across various tissues in *G. hirsutum*

To investigate the potential developmental roles of the *G. hirsutum EDS1* gene family, transcriptome data were retrieved from the Multi-Omics Database for Cotton (https://yanglab.hzau.edu.cn/CottonMD*)*. The expression patterns of *GhEDS1* genes were analysed across four major tissue categories: fibres (A), reproductive organs (B), vegetative tissues (C), and ovules (D), and visualised via heatmaps (Fig. 9A–D). In fibre tissues, several genes particularly *GhEDS1A-19*, *GhEDS1A-20*, *GhEDS1A-40*, *GhEDS1A-3*, and *GhEDS1D-48* showed consistently higher expression during 10–25 days post-anthesis (DPA), suggesting potential roles in fibre elongation or

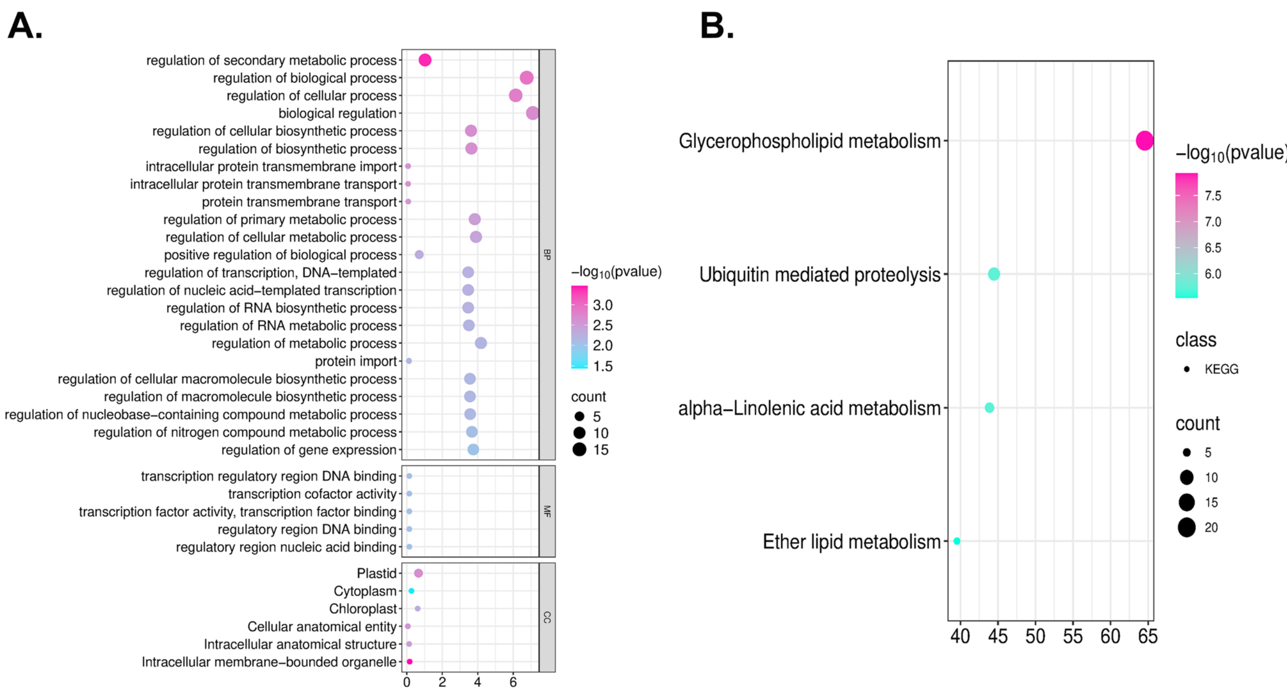

Fig. 8 Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analyses of GhEDS1 genes (**A**) Highly enriched GO terms in *GhEDS1* genes. **B** Highly enriched KEGG pathways in *GhEDS1* genes

secondary cell wall biosynthesis. In reproductive organs, including the anther and filament, *GhEDS1D-53* and *GhEDS1A-7* exhibited moderate to strong expression, whereas most other family members displayed weak or tissue-restricted expression. In pistils, genes such as *GhEDS1A-30* and *GhEDS1D-75* showed higher expression, potentially indicating functions related to fertilisation or pistil maturation. In vegetative tissues (root, stem, leaf), *GhEDS1A-3*, *GhEDS1A-30*, and *GhEDS1D-48* showed relatively broad and elevated expression levels. Overall, the non-redundant and tissue-specific expression profiles of *GhEDS1* genes suggest possible functional divergence and developmental specialisation within this gene family.

Expression profiling of *GhEDS1* genes in response to abiotic stress

To investigate the possible involvement of *GhEDS1* genes in abiotic stress responses, transcriptional dynamics of the predicted *GhEDS1* genes were further dissected using publicly available RNA-seq datasets of *G. hirsutum* (TM-1) exposed to cold (4 °C), heat (37 °C), salt (200 mM NaCl) and polyethylene glycol (PEG)-induced drought mimicking conditions (20% PEG6000). As shown in Fig. 10A–D, *GhEDS1* family members showed distinct and stress-specific expression patterns over multiple time points (1, 3, 6, 12 and 24 h).

Under cold stress (Fig. 10A), several genes, including *GhEDS1A-03*, *GhEDS1D-84* and *GhEDS1D-48*, showed strong induction at early to intermediate time points (3 h and 6 h), indicating a possible role in cold perception and signaling. In contrast, heat treatment (Fig. 10B) strongly upregulated *GhEDS1A-13* and *GhEDS1A-30* between 6 h and 12 h, while GhEDS1A-19 showed only modest induction, suggesting a selective transcriptional response under thermal stress. Salt stress (Fig. 10C) resulted in a marked increase in transcript levels for *GhEDS1D-48*, *GhEDS1A-03* and *GhEDS1A-13* during the early phases of stress (1 h to 6 h), indicating a possible role in the rapid response to ion toxicity and osmotic adjustment mechanisms. Notably, under PEG-induced osmotic stress (Fig. 10D), which mimics drought conditions, several GhEDS1 genes showed persistent expression changes. *GhEDS1A-19*, *GhEDS1D-75* and *GhEDS1D-65* were strongly upregulated at later time points (12 h and 24 h), suggesting that they are likely involved in prolonged adaptation to desiccation. In addition, *GhEDS1D-69* showed moderate induction during early exposure, suggesting a biphasic transcriptional response. This multi-level expression pattern suggests that certain *GhEDS1* genes may be involved in both early signaling events and downstream protective mechanisms during water deprivation. The sensitivity of these genes to PEG treatment highlights their functional importance in drought-related signaling pathways, particularly in the adaptation of cotton to osmotic imbalance. Interestingly, certain genes such as *GhEDS1A-07* and *GhEDS1D-04* remained weakly expressed or largely stress-independent, while others

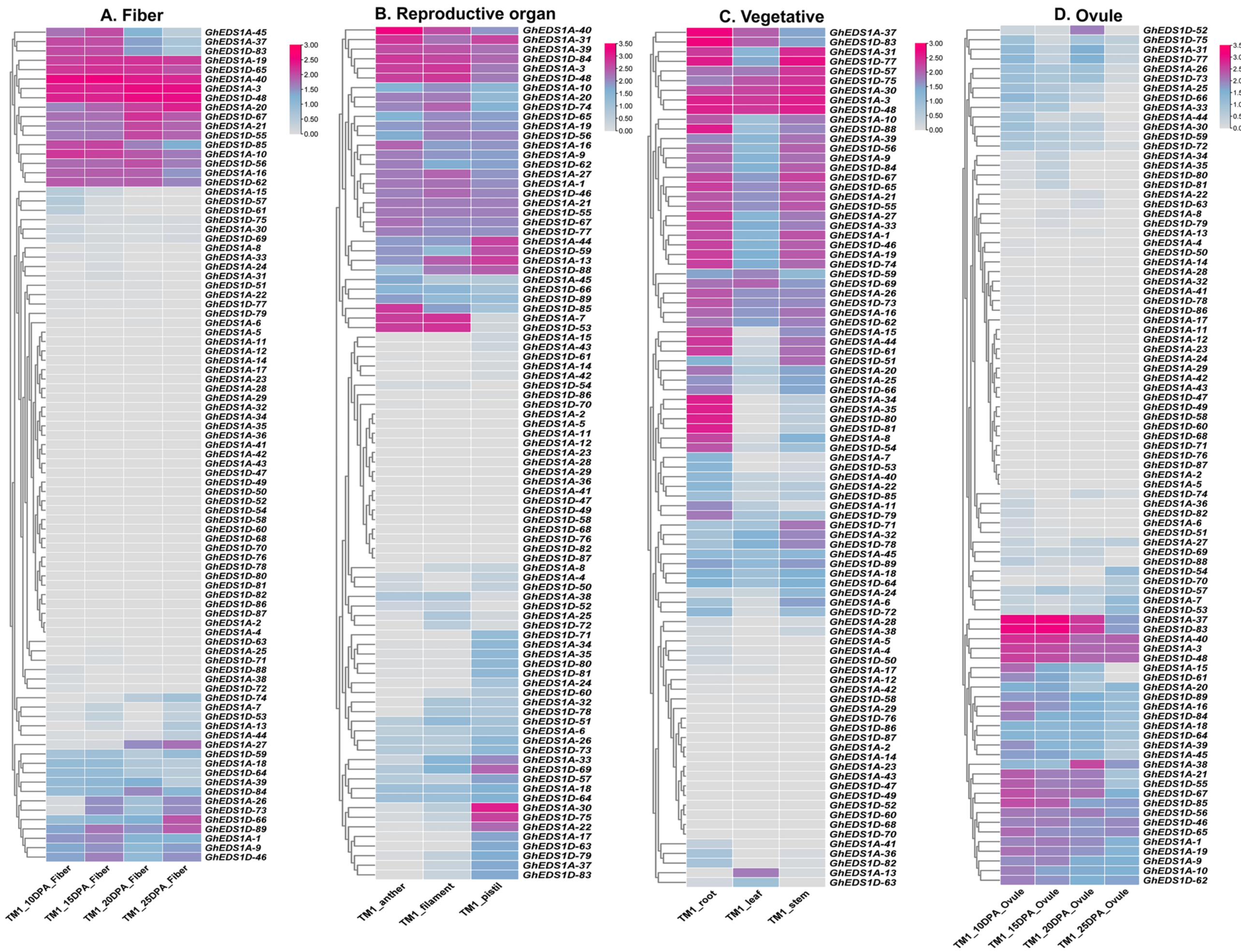

Fig. 9 Expression profile of 89 *GhEDS1* genes in upland cotton. A heat map showing the expression profile of GhEDS1 genes in different tissues of cotton (**A**) Fibre. **B** Reproductive organ, **C** Ovule, **D** Vegetative organs

showed specific induction under only one stress condition, supporting the notion of sub-functionalisation and specialisation among the *GhEDS1* paralogues.

To validate the RNA-seq data, qRT-PCR was performed on leaves of G. hirsutum (cv. Shayan) exposed to 20% PEG6000. *GhEDS1D-57* and *GhEDS1D-48* showed consistent upregulation at all time points, while *GhEDS1A-13* showed strong induction (~7-fold) at 12 h, indicating early osmotic stress sensitivity. *GhEDS1D-33* and *GhEDS1D-30* showed a gradual, time-dependent increase, confirming their functional significance for sustained stress adaptation. *GhEDS1A-30* showed a moderate induction that peaked after 12 h, consistent with the RNA-seq data. These results are illustrated in Fig. 11, which shows the qRT-PCR expression patterns of selected *GhEDS1* genes under PEG-induced osmotic stress, confirming their role in drought-responsive regulatory pathways. These patterns suggest distinct but coordinated functions of *GhEDS1* homologues in PEG-induced stress responses.

Discussion

The *EDS1* gene family comprises important intracellular immune regulators that serve as central nodes in pathogen-induced and abiotic stress-responsive signalling networks in plants [36]. These proteins possess two conserved domains an N-terminal α/β-hydrolase fold and a C-terminal EDS1–PAD4 (EP) domain that facilitate the formation of heterodimeric complexes with Phytoalexin Deficient 4 (PAD4) and Senescence-Associated Gene 101 (SAG101), which are essential for downstream immune signal transduction [9, 37]. In model species such as *Arabidopsis thaliana* and *Oryza sativa*, EDS1 coordinates effector-triggered immunity (ETI), reactive oxygen species (ROS)-mediated signalling, and hormonal crosstalk—particularly involving salicylic acid (SA) and abscisic acid (ABA)—to regulate a broad range of defence and stress responses [38–40]. However, despite extensive functional characterization in model plants, genome-wide analysis of EDS1 genes in polyploid cotton (Gossypium spp.), a crop of major agronomic importance, has

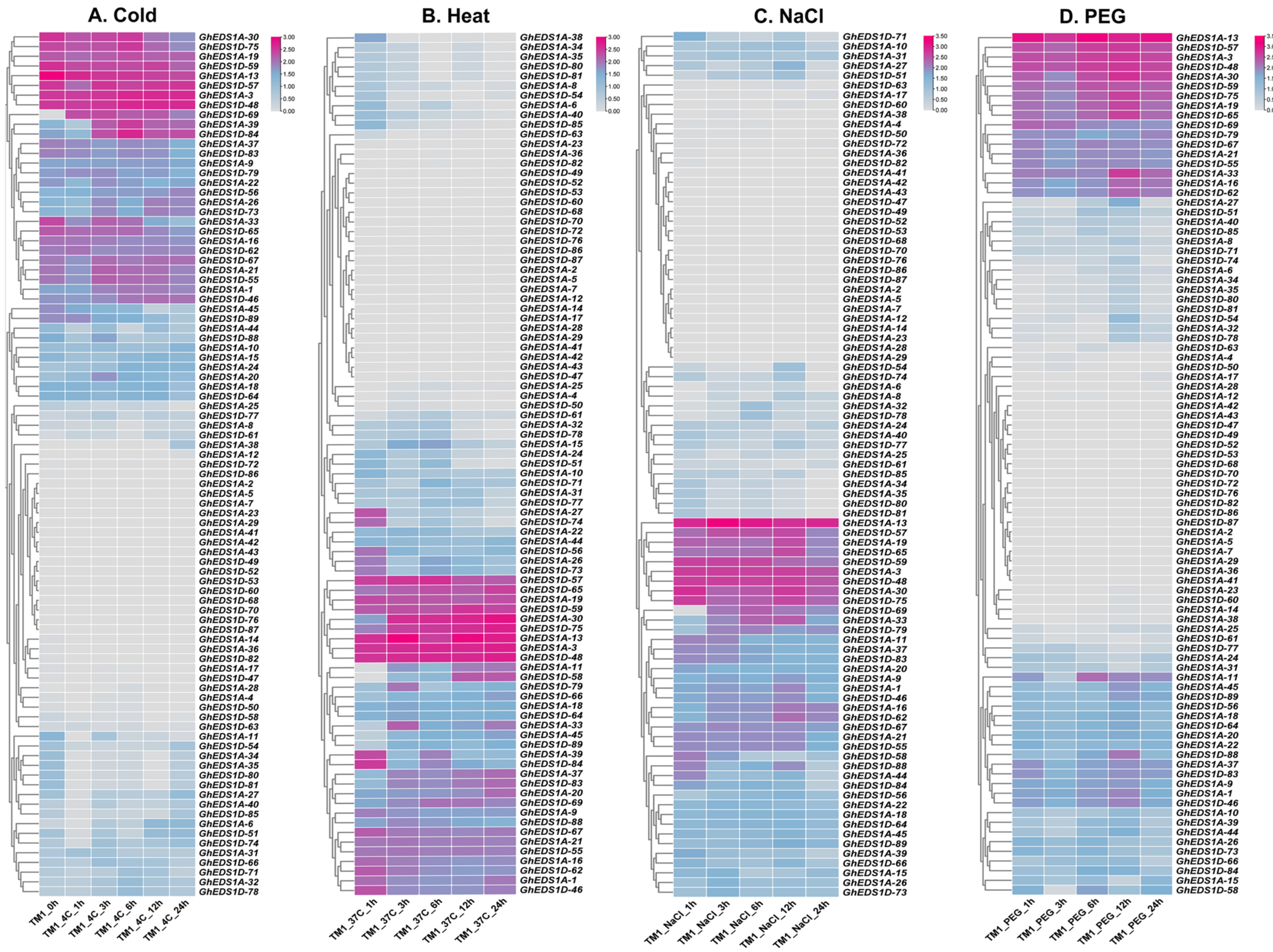

Fig. 10 Expression pattern analysis of *GhEDS1* genes in response to abiotic stress. **A** Cold, **B** Heat, **C** NACL, **D** PEG. FPKM values were log2 transformed and the heatmap was constructed using TBTools software. The red colour shows the highest and the blue colour shows the lowest expression levels in the expression bar

remained largely unstudied, particularly with respect to their role in integrating various stress factors.

Our genome-wide analysis identified 268 putative EDS1 genes in four *Gossypium* species: 89 in G. *hirsutum*, 86 in G. *barbadense*, 48 in G. *arboreum* and 45 in G. *raimondii*. This significant expansion in the tetraploid species far exceeds that in A. thaliana (4 genes) and *Glycine max* (11 genes) [41, 42]. This expansion likely reflects both ancient and recent whole genome duplications (WGD) and segmental duplication events [43]. Similar duplication-driven expansions have also been observed in other cotton gene families, including *CER*, *CKX* and *FORMIN*, highlighting the evolutionary pressure to maintain genes under stress-adaptive selection [4, 44, 45]. The extensive retention of duplicated paralogues in tetraploid cotton suggests a functional significance that may promote transcriptional versatility, subfunctionalisation or neofunctional innovation in stress-induced signalling pathways [46].

Chromosomal mapping revealed non-random aggregation of *GhEDS1* genes, particularly on chromosomes A11 and D11 in *G. hirsutum*, and A08, A10, and D11 in *G. barbadense*, suggesting the existence of genomic hotspots for EDS1 gene accumulation [47]. These regions may represent zones of high chromatin accessibility or sites of selective evolutionary retention linked to stress regulation [48]. Comparable clustering patterns have been reported in defence-related loci such as NBS-LRR gene clusters in *Solanum* and *Vitis*, where tandem and segmental duplications are believed to be driven by positive selection for functional diversification [49, 50]. Phylogenetic analysis grouped the cotton EDS1 protein sequences into six well-supported clades (GEDS1-I to GEDS1-VI), comprising orthologues from both diploid and tetraploid species, along with *A. thaliana*. The coexistence of conserved and clade-specific expansions suggests a dual evolutionary trajectory involving the retention of ancestral genes alongside lineage-specific divergence [51]. Interestingly, one *Arabidopsis* orthologue formed a distinct

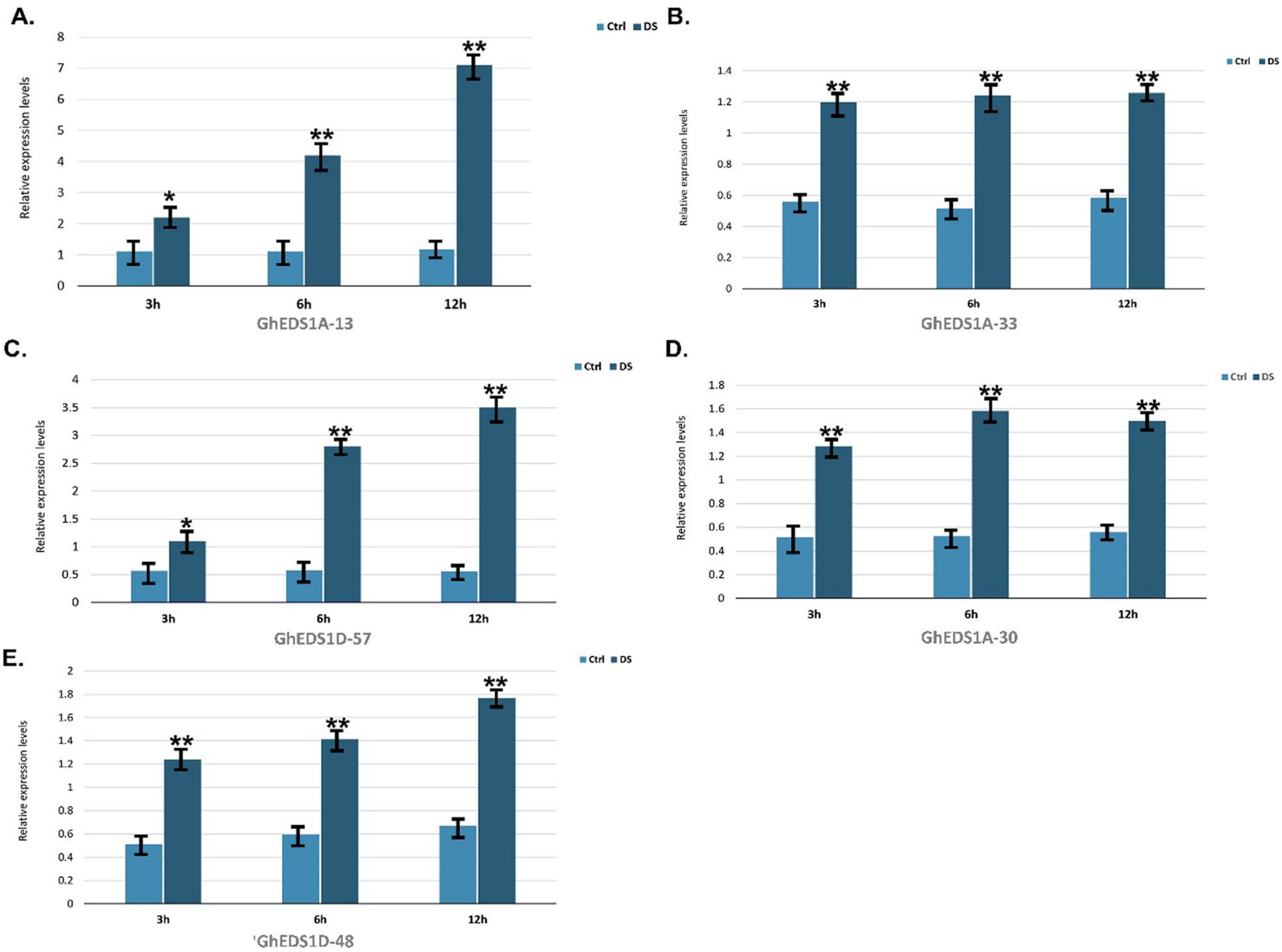

Fig. 11 Relative expressions of *GhEDS1* genes. qRT-PCR analysis was performed to observe the relative expression patterns of *GhEDS1* genes at 3 h, 6 h and 12 h under control and drought stress conditions. Vertical bars represent mean ± SD ($n = 3$). * and ** indicate significance at $p \leq 0.05$ and $p \leq 0.01$, respectively

outlier clade, potentially indicating species-specific neofunctionalisation or the absence of corresponding homologs in cotton [52]. By contrast, the remaining *Arabidopsis* EDS1 genes clustered with their cotton counterparts, highlighting conserved evolutionary relationships. Previous studies have shown that genes within the same phylogenetic clade often share similar expression profiles and functional roles, supporting the use of clade-based predictions in gene function annotation [53, 54]. The consistent pairing of homologous genes from the A and D subgenomes within clades further supports the notion of retained functional conservation following allopolyploidisation [55]. This observation aligns with findings in other polyploid systems, where homoeologous genes exhibit coordinated regulatory behaviour [14, 56].

The identification of conserved protein motifs within gene families provides key insights into evolutionary constraints, structural integrity, and potential functional divergence [57]. In our analysis, GhEDS1 proteins displayed considerable motif variability (1 to 15 motifs per protein), yet several motifs were consistently conserved across subfamilies, indicating their essential role in maintaining core biochemical functions. This conservation of central motifs, alongside variability in peripheral regions, suggests evolutionary pressure to preserve functional domains while allowing subfamily-specific specialisation [58–60]. Such structural modularity has also been reported in other gene families, where core motifs maintain activity and variable regions support interaction flexibility or regulatory adaptation [14]. In A. *thaliana*, EDS1 proteins similarly retain conserved EP and α/β-hydrolase domains critical for immune signalling, while divergent N- and C-terminal regions influence partner specificity and subcellular localization [8]. Although motif-level analyses of EDS1 proteins in other crops are limited, similar domain conservation with species-specific divergence has been observed in soybean, reflecting parallel evolutionary trajectories [41]. Together, these observations support a model of structural conservation underpinning functional stability, coupled with adaptive diversification that facilitates lineage-specific regulatory complexity.

Analysis of cis-regulatory elements in the promoter regions of *GhEDS1* genes revealed a diverse enrichment of hormone- and stress-responsive motifs, including ERE, ABRE, GARE, TCA, ARE, G-box, Box-4, and GATA elements. This pattern suggests transcriptional regulation by an intricate network of phytohormonal and environmental signals [61]. ABREs (abscisic acid-responsive elements), essential for ABA-mediated expression under drought and osmotic stress, were prevalent, indicating responsiveness to ABA-driven signalling [62]. GARE motifs, responsive to gibberellins, have been implicated in regulating growth and salt/drought adaptation [63], while TCA elements mediate salicylic acid (SA)-dependent responses linked to pathogen defence and systemic acquired resistance [64]. EREs (ethylene-responsive elements), often acting synergistically with AP2/ERF transcription factors, contribute to dehydration, salt, and cold stress responses [65]. The G-box (CACGTG), a multifunctional motif targeted by bZIP and bHLH TFs, plays a central role in ABA signalling, light responsiveness, and drought adaptation, frequently co-occurring with ABREs in stress-inducible promoters [66]. AREs and Box-4 elements imply regulation under hypoxic and light-responsive conditions, while GATA motifs further link ***GhEDS1*** transcription to light signalling, nitrogen metabolism, and defence priming [67]. Taken together, this cis-element landscape supports the role of *GhEDS1* genes as transcriptional integrators that mediate flexible responses to diverse hormonal and environmental stimuli in cotton.

Gene duplication is a key driver of gene family expansion and functional diversification in plants, primarily through whole-genome duplication (WGD), tandem duplication, and segmental duplication events [68]. The final size and structure of gene families are shaped not only by polyploidisation and genome fractionation, but also by post-duplication selection pressures that determine paralogue retention or loss [69]. In *Gossypium*, successive WGD events have led to a complex evolutionary landscape marked by both gene retention and lineage-specific gene loss. For instance, the subtilisin-like protease (SBT) and SAC gene families in *G. hirsutum* are reduced in size compared to their diploid progenitors *G. raimondii* and *G. arboreum*, reflecting gene contraction after polyploidisation [51, 70].

A similar trend was observed for the EDS1 gene family: *G. hirsutum* and *G. barbadense* harbour fewer GhEDS1 genes than the combined total of their diploid ancestors. This suggests post-polyploid gene loss, likely due to functional redundancy or relaxed purifying selection [71]. Such reduction is consistent with broader patterns in allotetraploid cotton, where duplicated genes often undergo pseudogenisation or transcriptional silencing owing to diminished selection pressure [72]. Nevertheless, our Ka/Ks analysis showed that over 90% of orthologous and paralogous GhEDS1 gene pairs exhibit Ka/Ks ratios < 1, indicating strong purifying selection. This selective constraint implies that retained EDS1 homologues fulfil essential roles and are evolutionarily conserved to preserve immune and stress-related functions [8]. This pattern—conservation despite contraction—is typical of diploidisation, wherein core regulatory genes are maintained to safeguard critical signalling pathways. Comparable evolutionary conservation of EDS1 genes has been reported in diverse plant lineages, including ***A. thaliana***, legumes, and cereals, underscoring their ancestral and indispensable roles in plant defence signalling [14, 41, 42]. The retention of specific EDS1 paralogues in cotton further highlights their functional importance and identifies them as promising targets for enhancing disease resistance and abiotic stress resilience through breeding programmes.

Protein–protein interaction (PPI) networks offer critical insights into the functional diversification and modular regulation of gene families [73]. In this study, the PPI network of GhEDS1 proteins, constructed via orthology with A. *thaliana*, revealed distinct interaction modules, indicating functional sub-compartmentalisation within the family. Rather than operating as a single defence hub, EDS1 proteins in cotton appear to be distributed across multiple coregulatory circuits—likely reflecting adaptation to polyploid genome dynamics and multifactorial stress environments [74]. *GhEDS1D-88* emerged as a central node, interacting with other *GhEDS1* homologues and phospholipase A (PLA1 and PLA2-ALPHA) family members, suggesting its involvement in lipid-mediated signalling pathways that intersect with hormonal and stress responses. This is notable given the growing recognition that lipid-derived molecules such as lysophospholipids and oxylipins act as secondary messengers in both immunity and abiotic stress adaptation [75–77]. These interactions imply a mechanistic link between membrane remodelling and EDS1-mediated signalling—an association not previously reported in cotton—indicating possible sub-functional roles beyond canonical immune receptor coordination.

Subnetworks involving *GhEDS1D-72*, *GhEDS1D-84*, and others indicate conserved interaction patterns alongside context-specific signalling divergence. This modular organisation parallels the functional flexibility of Arabidopsis EDS1–PAD4–SAG101 complexes, where subunit composition influences downstream outcomes such as SA-mediated defence and ET–JA crosstalk [8]. Such comparisons support the idea that GhEDS1 paralogues have evolved to partition functional roles, enabling more nuanced responses to diverse stress cues.

The identification of *Arabidopsis* orthologues such as *AT4G19520* and *T1K7.18*, known for roles in hormonal and immune signalling, underscores the conservation of stress-related pathways across dicot lineages [78]. Moreover, KEGG enrichment analysis highlighting pathways such as glycerophospholipid and α-linolenic acid metabolism further supports the integration of *GhEDS1* proteins into membrane-associated and lipid-responsive signalling networks. Collectively, these results support a model in which GhEDS1 proteins act not as isolated units but as context-specific signalling components shaped by lipid cues and subfamily-specific interactions. Such regulatory plasticity may be critical to cotton's ability to mount tailored responses to overlapping abiotic and biotic stresses.

Transcription factors (TFs) are important regulators of plant gene expression in response to environmental cues and developmental processes [79]. Our analysis identified 2,792 TFs potentially regulating 87 *GhEDS1* genes, with *Dof*, *AP2/ERF*, *MIKC_MADS* and *C2H2* being the most abundant families. This diversity suggests that GhEDS1 expression is controlled by intricate transcriptional networks related to stress and development. The Dof family, the most abundant in our dataset, is associated with redox regulation and abiotic stress responses [80]. *AP2/ERF* and *bZIP* TFs important regulators of desiccation, ethylene and ABA signalling were also highly represented, suggesting a possible role in the activation of *GhEDS1* under stress [81, 82]. Other TFs such as MYB, WRKY, TCP and NAC, which coordinate immune and hormonal responses, support the involvement of GhEDS1 in complex stress adaptation mechanisms. Notably, *GhEDS1A-1* and *GhEDS1D-46* were associated with more than 25 TF families, highlighting their potential role as regulatory hubs. Taken together, these results suggest that GhEDS1 genes are strongly modulated by different TFs to integrate immune, developmental and environmental signalling. Future research should validate key TF–GhEDS1 interactions by functional promoter assays.

Post-transcriptional regulation by microRNAs (miRNAs) also contributes to adaptation to environmental stress in plants [83]. Our predictions identified 85 high-probability *miRNA–GhEDS1* interactions involving 32 miRNAs, highlighting a complex regulatory level. Among them, the ghr-miR414 family was predominantly targeted *to GhEDS1A-21*, suggesting a regulatory hub in the drought and salinity response networks [84, 85]. Other important interactions included ghr-miR2950 (linked to antiviral defence) [86] and ghr-miR394 (involved in ABA signalling) [45], highlighting the dual role of *GhEDS1* in biotic and abiotic stress responses. Notably, miRNA-regulated genes such as *GhEDS1A-13* and GhEDS1D-19 were also significantly upregulated under PEG-induced drought stress, suggesting coordinated regulation at transcriptional and posttranscriptional levels. This overlap suggests that specific miRNAs may act as molecular switches to fine-tune GhEDS1 activity, allowing a stress response without compromising growth. Such plasticity reflects the adaptability of the EDS1 network in regulating osmotic stress.

Expression profiling revealed that *GhEDS1* genes exhibited dynamic, stress-responsive transcription patterns, particularly under PEG-induced osmotic stress (Fig. 10D). Several genes, including *GhEDS1A-65*, *GhEDS1D-19*, *GhEDS1D-48*, and *GhEDS1A-13*, were significantly upregulated, a pattern corroborated by qRT-PCR validation (Fig. 11). Notably, *GhEDS1A-08* showed sustained induction across all time points, peaking at ~ 7-fold after 24 h, suggesting a role in long-term desiccation tolerance, possibly via ABA signalling and oxidative stress mitigation. Similarly, the progressive induction of *GhEDS1A-33* and *GhEDS1A-30* points to their involvement in extended stress response cascades. These transcriptional trends align with the presence of stress-related cis-elements—such as MBS (drought-responsive), ABRE (ABA-responsive), and DRE (dehydration-responsive)—in their promoters (Fig. 5), supporting a role in ABA-dependent and ABA-independent regulatory pathways. In particular, *GhEDS1A-13* and *GhEDS1D-57*, enriched in ABRE motifs, may contribute to ABA-mediated responses including stomatal closure, ion homeostasis, and ROS detoxification. These findings are consistent with reports from *A. thaliana* and *G. max*, where *EDS1* enhances stress tolerance by modulating antioxidant activity and hormonal signalling [41, 87, 88]. Previous studies have also highlighted *AtEDS1* as a key integrator of biotic and abiotic stress signals via PAD4/SAG101 interactions and SA–ABA crosstalk [89, 90], a function likely conserved in cotton. Furthermore, our PPI and promoter TF network analyses reinforce the regulatory anchoring of *GhEDS1* genes within broader transcriptional frameworks. Promoter enrichment for NAC, DREB, bZIP, and WRKY binding sites—well-known for their roles in abiotic stress perception supports the hypothesis that ***GhEDS1*** genes act as regulatory hubs [91]. This is further substantiated by integrative evidence, including expression dynamics, cis-element enrichment, miRNA targeting, protein interactions, and pathway enrichment, offering a robust functional inference framework. Taken together, these data strongly support the roles of *GhEDS1A-13*, *GhEDS1D-48*, and *GhEDS1D-57* as key regulators not merely passive stress markers, but active components of drought-responsive signalling. Their dual involvement in ABA- and ROS-mediated pathways positions them as promising candidates for functional validation and genetic improvement through transgenics or genome editing in stress-resilient cotton breeding.

This study highlights both conserved and species-specific features of the EDS1 gene family across cotton and model systems such as A. *thaliana* and O. *sativa*, where EDS1 proteins play pivotal roles in immune signalling and stress integration. However, the comparative expansion, cis-regulatory complexity, and miRNA-mediated post-transcriptional control observed in cotton suggest a higher level of transcriptional plasticity, potentially shaped by polyploid evolution and environmental adaptation. From an applied perspective, our findings provide valuable insights into stress-responsive regulatory networks in cotton, with direct implications for breeding programmes. Notably, GhEDS1A-13, GhEDS1D-48, and GhEDS1D-57 emerge as robust functional candidates for genome editing due to their strong transcriptional activation under PEG-induced stress, enrichment for ABA-responsive and ROS-linked cis-elements, and regulatory targeting by stress-associated miRNAs (ghr-miR414, ghr-miR394). These genes represent ideal targets for CRISPR/Cas9-mediated editing or overexpression strategies aimed at enhancing drought resilience in upland cotton.

Moreover, the identification of 85 putative miRNA–GhEDS1 interactions many of which involve miRNAs not previously reported to regulate EDS1 genes in cotton constitutes a significant advancement. To our knowledge, this is the first report elucidating such an extensive miRNA–GhEDS1 network in the context of abiotic stress, offering novel insights into post-transcriptional regulatory mechanisms. These findings warrant further validation through degradome sequencing, miRNA–target reporter assays, and transgenic functional studies. Our results are in line with recent reports highlighting key regulatory genes that improve fibre traits and stress resilience in cotton [92, 93], and further reinforce the evolutionary and functional importance of stress-modulatory genes in modern cotton breeding [94, 95]. Collectively, this work lays a foundational framework for future functional genomics and marker-assisted breeding aimed at enhancing stress tolerance in Gossypium species.

Conclusion

In this study, we performed the first genome-wide characterisation of the *GhEDS1* gene family in four Gossypium species. We identified 268 members and uncovered their structural diversity, evolutionary history and role as stress receptors. Segmental duplications and whole genome duplications were the main drivers of expansion, with most paralogues undergoing purifying selection. Promoter and network analyses revealed extensive regulatory complexity, including interactions with stress-related transcription factors and miRNAs, in particular ghr-miR414. Expression profiling and qRT-PCR validation identified *GhEDS1A-13*, *GhEDS1D-57* and *GhEDS1D-48* as key drought-responsive genes, suggesting a role in adaptation to osmotic stress. These results provide a mechanistic basis for improving stress resistance in cotton through molecular breeding. In particular, the identified GhEDS1 candidates can be utilised in future gene function studies and transgenic experiments to evaluate their role under field-relevant drought conditions. In addition, the discovery of miRNA-mediated post-transcriptional regulation, particularly involving *ghr-miR4*, opens up new possibilities for the design of synthetic regulatory modules and the development of stress-inducible promoters. Given the increasing accessibility of genome editing tools such as CRISPR/Cas9, precise manipulation of *GhEDS1* genes or their regulatory elements offers promising strategies to improve abiotic stress tolerance in elite cotton cultivars. Future research should focus on the validation of these targets through functional assays, transgenic models and field-level phenotyping, as well as breeding integration through Agrobacterium-mediated transfer or pollen tube transformation systems. Overall, this study establishes a genomic and functional framework for translating insights from the *GhEDS1* gene family into targeted biotechnological applications to improve stress resistance in cotton.

Supplementary information

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Supplementary Material 1

Supplementary Material 2

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Data availability
All relevant data are provided within the manuscript and the supplementary information files.

Declarations

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All experimental studies on plants were conducted in compliance with relevant institutional, national, and international guidelines and legislation.

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